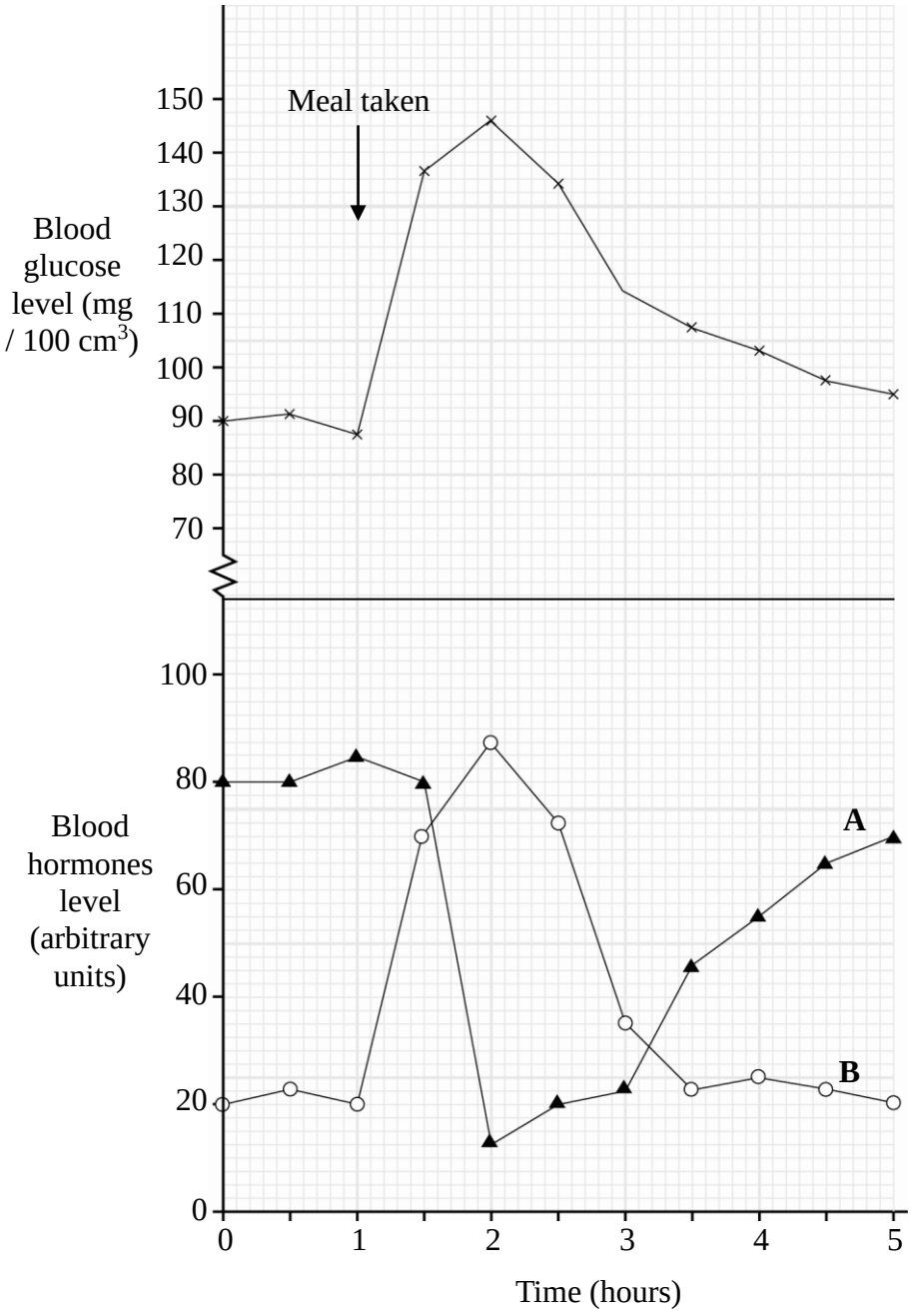
Short questions

|!|BCESB0190000001|!| ★

Two hormones, insulin and glucagon, are involved in the regulation of the blood glucose level. In an investigation, the blood glucose, insulin and glucagon levels were measured in a group of people over 5 hours. They ingested a carbohydrate-rich meal after the first hour. The results are shown in the following graphs.



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(a) Based on the graphs, identify what hormones A and B are. (2 marks)

(b) Describe the mechanism that leads to the changes in the blood glucose level after the ingestion of a carbohydrate-rich meal. (3 marks)

##

(a) A: glucagon (1)
B: insulin (1)

(b) Carbohydrates are broken down and glucose is absorbed in the alimentary canal. This raises the blood glucose level. (1)
Pancreas detects a high blood glucose level and secretes more insulin and less glucagon into the blood. (1)
Insulin stimulates liver cells to convert excess glucose to glycogen which is stored in the liver and muscles. (OR Insulin stimulates the uptake of glucose by body cells and subsequent oxidation of glucose by respiration.) (1)
As a result, the blood glucose level falls.

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	3

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|!|BCESC0190000002|!|

For each of the parameters in the human body listed in ***Column 1***, select from ***Column 2*** one description that matches it. Put the appropriate letter in the space provided.

(4 marks)

Column 1		***Column 2***
Water content in blood	(a) _____	A. regulated by the liver and the pancreas
Body temperature	(b) _____	B. regulated by the stomach
Glucose level in blood	(c) _____	C. regulated by the kidneys
Gas content in blood	(d) _____	D. regulated by controlling breathing and heartbeat
		E. mainly regulated by the skin

##

(a) C (1)
(b) E (1)
(c) A (1)
(d) D (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESC0190000003|!|

Insulin is a hormone secreted by the pancreas. The amount of insulin secreted varies under different conditions.

(a) What is the chemical nature of insulin? (1 mark)

_____

(b) State ***two*** responses stimulated by insulin. (2 marks)

_____

_____

_____

_____

##

(a) It is protein. (1)

(b) Insulin stimulates the conversion of glucose to glycogen by liver cells. (1)
Insulin stimulates the uptake of glucose by body cells and subsequent oxidation of glucose by respiration. (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2

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